Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 2 — Coordinate geometry in the (x, y) plane
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第2章 · 圆的坐标几何
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$.

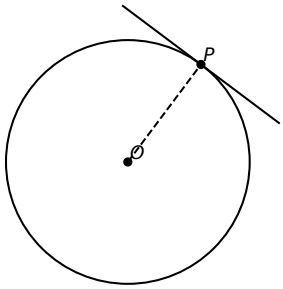


Figure 1

(a) Find its centre and radius. (3)

(b) Find the equation of the tangent at $P = (6, 2)$. (3)

(Total for Question 1 is 6 marks)

$A = (-2, 1)$ and $B = (6, 5)$ are endpoints of a diameter of a circle.

(a) Find an equation of the circle. (3)

(b) Find where the circle meets the line $y = 3$. (2)

(Total for Question 2 is 5 marks)

A circle has centre $(1, 4)$ and passes through $(5, 7)$.

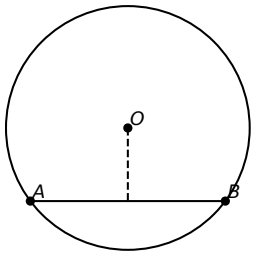


Figure 3

(a) Find its equation. (2)

(b) Find the exact length of the chord cut off by $y = 1$. (3)

(Total for Question 3 is 5 marks)

4.

The circle is $(x+2)^2+(y-1)^2=16$.

(a) Find the intersections with the line $x=0$. (2)

(b) Find equations of both tangents parallel to the y-axis. (2)

(Total for Question 4 is 4 marks)

5.

The line $y = x + k$ is tangent to $x^2 + y^2 = 18$.

(a) Find both possible values of k. (3)

(b) Find the point of contact for $k = 6$. (2)

(Total for Question 5 is 5 marks)

6.

A circle has equation $x^2 + y^2 + 2gx + 2fy = 0$. It passes through $(4, 0)$ and $(0, 6)$.

(a) Find g, f and the centre. (3)

(b) Show that the origin and $(4, 6)$ are opposite ends of a diameter. (3)

(Total for Question 6 is 6 marks)

The circle C is $(x-3)^2+(y+1)^2=25$, and $P=(9,7)$.

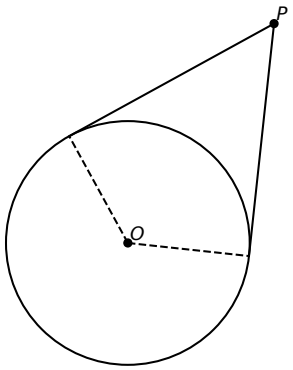


Figure 7

(a) Determine whether P is inside, on or outside C. (2)

(b) Find the exact length of a tangent from P to C. (3)

(Total for Question 7 is 5 marks)

8. P2-C02-R08 · P2 · Chapter 2

A circle has centre $O = (2, -1)$. A chord has endpoints $A = (-1, 3)$ and $B = (5, 3)$.

(a) Find the radius of the circle. (2)

(b) Find the midpoint of AB and verify that the line from O to this midpoint is perpendicular to AB. (3)

(Total for Question 8 is 5 marks)

The circles are $C_1 : x^2 + y^2 = 25$ and $C_2 : (x - 6)^2 + y^2 = 13$.

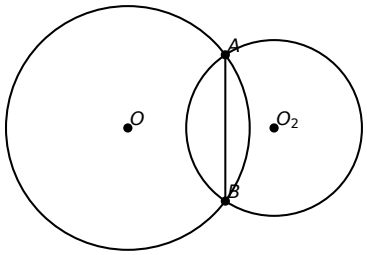


Figure 9

(a) Show that their common chord lies on $x = 4$. (2)

(b) Find the intersection points and the length of the common chord. (3)

(Total for Question 9 is 5 marks)

C is $x^2 + y^2 - 8x - 6y + 16 = 0$.

(a) Find its centre and radius. (3)

(b) Find all tangents to C parallel to the x-axis, including their contact points. (3)

(Total for Question 10 is 6 marks)

The circle $(x - a)^2 + y^2 = 25$ passes through $(1, 3)$. It is given that $a > 0$.

(a) Find a. (3)

(b) Find the equation of the tangent at $(1, 3)$. (3)

(Total for Question 11 is 6 marks)

The line $y = 2$ cuts the circle $x^2 + y^2 = 20$ at A, B.

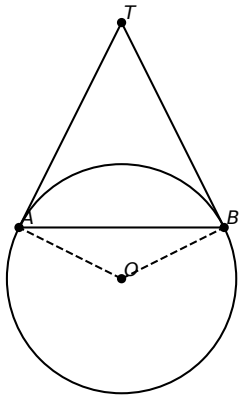


Figure 12

(a) Find A, B and the length AB. (3)

(b) Tangents at A, B meet at T. Find T. (3)

(Total for Question 12 is 6 marks)

13.

A circle has centre $(h, 2)$ and is tangent to the y-axis. It passes through $(4, 4)$, and $h > 0$.

(a) Find h and its radius. (3)

(b) Find its other intersection with the horizontal line $y = 4$. (2)

(Total for Question 13 is 5 marks)

14.

The circle C has centre $(2, 3)$ and is tangent to $y = 7$.

(a) Find an equation of C. (2)

(b) Find the exact coordinates where C meets $y = x + 1$. (3)

(Total for Question 14 is 5 marks)

15.

A circle has diameter endpoints $A = (1, 2)$ and $B = (7, 10)$.

(a) Find its equation and determine whether $P = (0, 6)$ lies on it. (3)

(b) Find an equation of the tangent at A. (3)

(Total for Question 15 is 6 marks)

16.

The circle C is $(x+1)^2+(y-2)^2=10$. The line L is $y=x+4$.

(a) Find both points where L meets C. (3)

(b) Find the midpoint of the chord and show that it lies on the perpendicular from the centre to L. (3)

(Total for Question 16 is 6 marks)

17.

A circle with centre $(3, -2)$ is tangent to the line $x = 8$.

(a) Find its equation and the point of contact. (3)

(b) Find the exact length of the chord on the line $y = 1$. (3)

(Total for Question 17 is 6 marks)

18. P2-C02-R18 · P2 · Chapter 2

The circle is $x^2 + y^2 + 4x - 10y + k = 0$. Its radius is 6.

(a) Find k and the centre. (3)

(b) Determine whether the circle cuts the x-axis. Find any intersections exactly. (3)

(Total for Question 18 is 6 marks)

The line $y = mx$ is tangent to $(x - 5)^2 + y^2 = 9$.

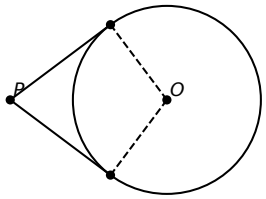


Figure 19

(a) Find the two possible values of m. (3)

(b) Find both contact points. (3)

(Total for Question 19 is 6 marks)

$A = (2, 0)$ and $B = (0, 4)$ lie on a circle whose centre lies on $y = x$.

(a) Find the centre. (3)

(b) Find the circle equation and its intersections with $y = x$. (4)

(Total for Question 20 is 7 marks)

21. P2-C02-R21 · P2 · Chapter 2

The circles $C_1 : (x-1)^2 + y^2 = 9$ and $C_2 : (x-6)^2 + y^2 = 4$ are given.

(a) Show that the circles touch externally. (3)

(b) Find the contact point and the common tangent there. (3)

(Total for Question 21 is 6 marks)

Circle C has centre $(0, 0)$ and radius 5. Point $P = (4, 3)$ lies on it.

(a) Find the tangent L at P. (3)

(b) Find the area of the triangle enclosed by L and the coordinate axes. (3)

(Total for Question 22 is 6 marks)

The circle C is $(x-2)^2+(y+3)^2=13$.

(a) Find the points on C whose x-coordinate is 4. (2)

(b) Find the equations of the tangents at these points and their intersection. (4)

(Total for Question 23 is 6 marks)

A circle has centre $(1, 2)$ and is tangent to the line $y = x + 6$.

(a) Find the foot of the perpendicular from the centre to the line. (3)

(b) Find an equation of the circle. (2)

(Total for Question 24 is 5 marks)

The circle C has diameter endpoints $A = (-1, 2)$ and $B = (5, 10)$. Point $P = (6, 3)$ is given.

(a) Find the equation of C. (3)

(b) Show that P lies on C and that angle APB is 90° using gradients. (3)

(Total for Question 25 is 6 marks)

The circle C is $x^2 + y^2 - 2x - 8y + 8 = 0$. The point $A = (1, 4)$ is given.

(a) Find the centre and radius and explain why A is not on C. (3)

(b) Find all points on C lying on the line $y = 2x + 2$. (3)

(Total for Question 26 is 6 marks)

The circle C is $x^2 + y^2 = 50$. The chord AB has midpoint $M = (3, 1)$.

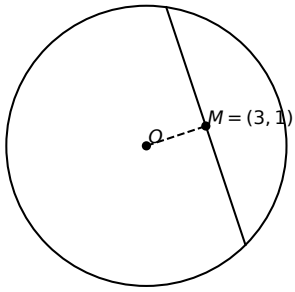


Figure 27

(a) Find an equation of the chord. (3)

(b) Find the exact chord length. (4)

(Total for Question 27 is 7 marks)

28.

A circle has equation $x^2 + y^2 + Dx + Ey + F = 0$ and passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$.

(a) Find D, E, F and the equation in centre-radius form. (4)

(b) Find the tangent at the origin. (3)

(Total for Question 28 is 7 marks)

29. P2-C02-R29 · P2 · Chapter 2

The line $x + y = k$ meets the circle $(x - 2)^2 + (y - 2)^2 = 8$.

(a) Find both values of k for which the line is tangent. (3)

(b) For $k = 4$, find the endpoints of the chord and its length. (4)

(Total for Question 29 is 7 marks)

Two concentric circles have centre $(2, 1)$. A line $y = 4$ cuts the outer circle in a chord of length 8 and is tangent to the inner circle.

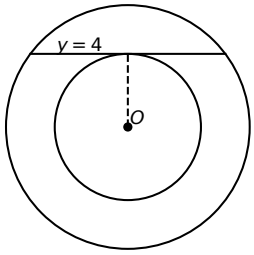


Figure 30

(a) Find equations of the two circles. (3)

(b) Find the exact area between the two circles. (2)

(Total for Question 30 is 5 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C02-R01 · 6 marks

(a) · 3 marks

Centre $(3, -2)$, radius 5.

配方得$(x-3)^2+(y+2)^2=25$。

M1	两坐标配方
A1	中心
A1	半径

(b) · 3 marks

$3x + 4y = 26$.

半径向量$(3,4)$，斜率$4/3$，切线$-3/4$，过P整理。

M1	半径梯度
M1	垂直切线过点
A1	方程

Question 2

P2-C02-R02 · 5 marks

(a) · 3 marks

$(x-2)^2+(y-3)^2=20.$

中心为中点(2,3)，半径平方$AB^2/4=(64+16)/4=20$。

M1	中心
M1	半径平方
A1	圆方程

(b) · 2 marks

$(2-2\sqrt{5},3),(2+2\sqrt{5},3).$

线通过中心，代y=3得$(x-2)^2=20$。

M1	正确代入
A1	两交点

Question 3

P2-C02-R03 · 5 marks

(a) · 2 marks

$(x-1)^2+(y-4)^2=25.$

半径平方$4^2+3^2=25$。

M1	两点距离平方
A1	方程

(b) · 3 marks

8.

代$y=1$得$(x-1)^2=16$，端点$x=-3,5$，同高距离8。

M1	联立求两横坐标
M1	取横差
A1	弦长

Question 4

P2-C02-R04 · 4 marks

(a) · 2 marks

$(0, 1 \pm 2\sqrt{3})$.

代x=0后$(y-1)^2=12$。

M1	正确平方关系
A1	两坐标

(b) · 2 marks

$x = -6$ and $x = 2$.

竖直切线位于中心横坐标±半径，即-2 ± 4。

M1	使用圆的水平极端位置
A1	两竖直方程

Question 5

P2-C02-R05 · 5 marks

(a) · 3 marks

$k = \pm 6$.

代入圆得$2x^2+2kx+k^2-18=0$；判别式$4k^2-8(k^2-18)=0$。

M1	代线入圆
M1	判别式等零
A1	两k

(b) · 2 marks

$(-3, 3)$.

相切二次式重根$x=-k/2=-3$，$y=x+6=3$。

M1	求重根
A1	接触点

Question 6

P2-C02-R06 · 6 marks

(a) · 3 marks

$g=-2, f=-3$; centre $(2,3)$.

点(4,0)给16+8g=0；(0,6)给36+12f=0，中心为(−g,−f)。

M1	两点代入
A1	两参数
B1	中心

(b) · 3 marks

Their midpoint is $(2,3)$ and both lie on the circle.

原点由圆式常数0可见在圆上；(4,6)代入16+36−16−36=0；中点为中心。

M1	检查另一点在圆上
M1	计算中点
A1	直径结论

Question 7

P2-C02-R07 · 5 marks

(a) · 2 marks

Outside, since $CP^2 = 6^2 + 8^2 = 100 > 25$.

比较点到中心距离平方与半径平方，无需先开方。

M1	正确距离平方
A1	分类与比较

(b) · 3 marks

$5\sqrt{3}$.

切点处半径垂直切线，勾股长度 $\sqrt{(CP^2 - r^2)} = \sqrt{75}$。

M1	使用直角切线三角形
M1	正确平方差
A1	根式

Question 8

P2-C02-R08 · 5 marks

(a) · 2 marks

5.

OA坐标差$-3,4$，长度5。

M1	半径两点距离
A1	半径

(b) · 3 marks

Midpoint $(2,3)$; AB horizontal and the centre-to-midpoint line is $x=2$, vertical.

先取中点，再比较水平弦和竖直中垂线，不能用无定义的竖线斜率相乘。

M1	中点
M1	两方向
A1	垂直验证

Question 9

P2-C02-R09 · 5 marks

(a) · 2 marks

Subtracting gives $-12x + 36 = -12$, so $x = 4$.

展开第二圆再与第一圆相减，消去x^2+y^2。

M1	展开并相减
A1	公共弦所在直线

(b) · 3 marks

$(4, 3), (4, -3)$; length 6.

$x=4$代第一圆得$y^2=9$，弦竖直用纵差6。

M1	回代解两个y
A1	两交点
A1	弦长

Question 10

P2-C02-R10 · 6 marks

(a) · 3 marks

Centre $(4, 3)$; radius 3.

配方得$(x-4)^2+(y-3)^2=9$。

M1	完成配方
A1	中心
A1	半径

(b) · 3 marks

$y = 0$ at $(4, 0)$ and $y = 6$ at $(4, 6)$.

水平切线在中心正上方和正下方，各相距$r=3$。

M1	正确极端位置
A1	两线
A1	接触点

Question 11

P2-C02-R11 · 6 marks

(a) · 3 marks

$a = 5.$

$(1-a)^2=16$给a=5或−3；正值限制选5。

M1	代点解平方方程
M1	使用正值限制
A1	a

(b) · 3 marks

$4x - 3y = -5.$

半径从(5,0)到(1,3)斜率−3/4，切线4/3；$y-3=(4/3)(x-1)$。

M1	半径斜率
M1	切线负倒数
A1	方程

Question 12

P2-C02-R12 · 6 marks

(a) · 3 marks

$(\pm 4, 2)$; $AB = 8$.

$x^2+4=20$，$x=\pm 4$，弦水平。

M1	联立
A1	两点
A1	长度

(b) · 3 marks

$T = (0, 10)$.

切线为$4x+2y=20$和$-4x+2y=20$，联立$x=0$,$y=10$；这些方程可由半径垂直推得。

M1	两条正确切线
M1	联立
A1	T

Question 13

P2-C02-R13 · 5 marks

(a) · 3 marks

$h = 5/2$; radius $5/2$.

到y轴的距离为h，故r=h；$(4-h)^2+4=h^2$给$20-8h=0$。

M1	切y轴的半径关系
M1	点条件
A1	中心参数和半径

(b) · 2 marks

$(1, 4)$.

关于中心竖直线x=5/2对称，两个交点横坐标和5；另一x=1，或联立求根。

M1	对称或圆线联立
A1	另一点

Question 14

P2-C02-R14 · 5 marks

(a) · 2 marks

$(x-2)^2+(y-3)^2=16$.

中心到水平切线距离7−3=4，即半径。

M1	半径
A1	方程

(b) · 3 marks

$(2\pm2\sqrt{2},3\pm2\sqrt{2})$, with matching signs.

直线过中心，代入得$2(x-2)^2=16$，$y=x+1$。

M1	正确联立简化
M1	求两根
A1	两配对点

Question 15

P2-C02-R15 · 6 marks

(a) · 3 marks

$(x-4)^2+(y-6)^2=25$; P gives $16\neq 25$, so P does not lie on it.

中心(4,6)，r^2=25；给定点(0,6)距中心4并非5。

M1	由直径求圆
M1	检查点
A1	正确结论

(b) · 3 marks

$3x+4y=11$.

半径斜率4/3，切线−3/4，过(1,2)得3x+4y=11。

M1	半径方向
M1	切线过点
A1	方程

Question 16

P2-C02-R16 · 6 marks

(a) · 3 marks

$\left(\frac{-3-\sqrt{19}}{2}, \frac{5-\sqrt{19}}{2}\right)$ and $\left(\frac{-3+\sqrt{19}}{2}, \frac{5+\sqrt{19}}{2}\right)$.

代线入圆得$(x+1)^2+(x+2)^2=10$，即$2x^2+6x-5=0$；二次公式给$x=(-3\pm\sqrt{19})/2$，回代$y=x+4$。

M1	代入建立二次方程
M1	解方程
A1	两点

(b) · 3 marks

Midpoint $(-3/2, 5/2)$, on $y=-x+1$.

二次方程两根平均为$-3/2$，直线给$y=5/2$；中心$(-1,2)$向斜率1直线的垂线为$y=-x+1$。

M1	中点
M1	中心垂线
A1	验证

Question 17

P2-C02-R17 · 6 marks

(a) · 3 marks

$(x-3)^2+(y+2)^2=25$; contact $(8,-2)$.

半径为8−3=5，接触点与中心同纵坐标。

M1	半径
A1	方程
B1	接触点

(b) · 3 marks

8.

距中心竖直3，半弦 $\sqrt{(25-9)}=4$，整弦8。

M1	正确中心距与半弦勾股
M1	从半弦到整弦
A1	长度

Question 18

P2-C02-R18 · 6 marks

(a) · 3 marks

$k = -7$; centre $(-2, 5)$.

配方得$(x+2)^2+(y-5)^2=29-k$；$29-k=36$给$k=-7$。

M1	正确配方
M1	半径条件
A1	k和中心

(b) · 3 marks

It cuts at $(-2 \pm \sqrt{11}, 0)$.

设$y=0$得$(x+2)^2=11>0$，故有两个不同实交点。

M1	联立x轴
M1	判断两实根
A1	两坐标

Question 19

P2-C02-R19 · 6 marks

(a) · 3 marks

$m = \pm 3/4$.

代入得$(1+m^2)x^2-10x+16=0$，判别式$100-64(1+m^2)=0$。

M1	圆线联立
M1	重根条件
A1	两斜率

(b) · 3 marks

$(16/5, 12/5)$ and $(16/5, -12/5)$.

重根$x=10/[2(1+9/16)]=16/5$；$y=m\,x$。

M1	重根横坐标
M1	对应纵坐标
A1	两点

Question 20

P2-C02-R20 · 7 marks

(a) · 3 marks

$(3,3)$.

设中心(h,h)，$OA^2=(h-2)^2+h^2$，$OB^2=h^2+(h-4)^2$；相等得h=3。

M1	中心参数化
M1	两半径平方相等
A1	中心

(b) · 4 marks

$(x-3)^2+(y-3)^2=10$; $(3\pm\sqrt{5},3\pm\sqrt{5})$.

$r^2=1+9=10$；联立y=x得$2(x-3)^2=10$。

M1	半径平方
A1	圆方程
M1	联立过心线
A1	两点

Question 21

P2-C02-R21 · 6 marks

(a) · 3 marks

Centre distance 5 equals the sum of radii $3+2$.

两中心在同一水平线上，距离6−1=5，与半径和相等，所以外切。

M1	中心距与两半径
M1	比较半径和
A1	外切结论

(b) · 3 marks

Contact $(4,0)$; tangent $x=4$.

接触点在连心线上，距第一中心向右3；半径水平所以切线竖直。

M1	定位接触点
A1	点
A1	切线

Question 22

P2-C02-R22 · 6 marks

(a) · 3 marks

$4x + 3y = 25$.

半径斜率3/4，切线−4/3；过(4,3)。

M1	正确垂直斜率
M1	过点
A1	方程

(b) · 3 marks

$625/24$ square units.

截距25/4和25/3，面积为其积的一半。

M1	两截距
M1	正确面积式
A1	精确面积

Question 23

P2-C02-R23 · 6 marks

(a) · 2 marks

$(4, 0)$ and $(4, -6)$.

代x=4得$(y+3)^2=9$。

M1	正确平方式
A1	两点

(b) · 4 marks

$2x + 3y = 8$ and $2x - 3y = 26$; intersection $(17/2, -3)$.

半径向量分别(2,3)、(2,−3)，切线用垂直关系；两线相加4x=34。

M1	第一切线
M1	第二切线
M1	联立
A1	交点

Question 24

P2-C02-R24 · 5 marks

(a) · 3 marks

$(-3/2, 9/2)$.

垂线斜率-1，过中心为y=−x+3，与y=x+6联立。

M1	中心垂线
M1	联立
A1	垂足

(b) · 2 marks

$(x-1)^2+(y-2)^2=25/2$.

半径平方=$(-5/2)^2+(5/2)^2=25/2$。

M1	垂足到中心距离平方
A1	圆方程

Question 25

P2-C02-R25 · 6 marks

(a) · 3 marks

$(x-2)^2+(y-6)^2=25$.

中点(2,6)，半径长度5。

M1	直径中点
M1	半径
A1	方程

(b) · 3 marks

$(6-2)^2+(3-6)^2=25$; $m_{PA}=1/7$, $m_{PB}=-7$, product -1.

先代圆式验证点，再计算过P的两边斜率，乘积−1。

M1	点在圆上验证
M1	两个斜率
A1	垂直结论

Question 26

P2-C02-R26 · 6 marks

(a) · 3 marks

Centre $A = (1, 4)$; radius 3; the centre is not on a circle of positive radius.

配方为$(x-1)^2+(y-4)^2=9$，A恰为中心而非周界点。

M1	配方
A1	中心与半径
B1	排除A在圆上的理由

(b) · 3 marks

$(1 \pm 3/\sqrt{5}, 4 \pm 6/\sqrt{5})$, with matching signs.

直线过中心，用$u=x-1$得$y-4=2u$，圆上$5u^2=9$。

M1	正确联立简化
M1	解正负u
A1	两点

Question 27

P2-C02-R27 · 7 marks

(a) · 3 marks

$3x + y = 10$.

OM斜率1/3，弦垂直OM故斜率−3且过M，y−1=−3(x−3)。

M1	弦与中心连线垂直
M1	用中点过线
A1	方程

(b) · 4 marks

$4\sqrt{10}$.

$OM^2=10$，半弦平方$r^2-OM^2=40$，整弦$2\sqrt{40}=4\sqrt{10}$。

M1	中心距平方
M1	半弦勾股
M1	乘2
A1	最简长度

Question 28

P2-C02-R28 · 7 marks

(a) · 4 marks

$D=-6, E=-8, F=0$; $(x-3)^2+(y-4)^2=25$.

原点给F=0，两个轴点给D=−6,E=−8，再配方。

B1	F
M1	求D和E
M1	配方
A1	标准式

(b) · 3 marks

$3x+4y=0$.

半径从中心(3,4)至原点，斜率4/3；切线−3/4且过原点。

M1	半径方向
M1	正确垂线
A1	方程

Question 29

P2-C02-R29 · 7 marks

(a) · 3 marks

$k = 0, 8.$

代y=k−x，或由中心向法线方向的距离平方$(k-4)^2/2=8$，得$(k-4)^2=16$。

M1	正确联立或垂距
M1	相切条件
A1	两k值

(b) · 4 marks

$(0, 4), (4, 0)$; length $4\sqrt{2}$.

线过中心，代y=4−x得$2(x-2)^2=8$，x=0,4，距离勾股。

M1	联立求端点
A1	两点
M1	两点距离
A1	弦长

Question 30

P2-C02-R30 · 5 marks

(a) · 3 marks

Outer $(x-2)^2+(y-1)^2=25$; inner $(x-2)^2+(y-1)^2=9$.

中心距线3，内半径3；外半弦4，勾股外半径5。

M1	内半径
M1	外半弦与勾股
A1	两方程

(b) · 2 marks

16π square units.

环面积π(25−9)，不能相减半径后再平方。

M1	两圆面积差
A1	精确面积